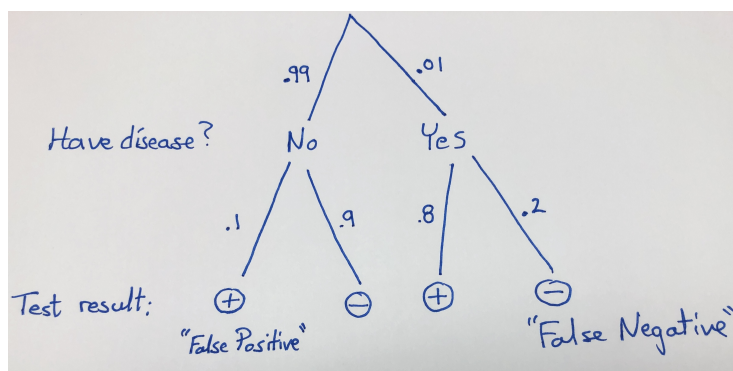


11 2026-02-06 | Week 04 | Lecture 11

11.1 Application of the law of total probability: diagnostic tests

This based on section 2.4

Example 40 (Bayes' Theorem). Prevalance of a disease in a population is 1%. A new diagnostic test is advertised as having a false positive rate of .1 and a false negative rate of .2. (In other words, 10% of positive tests are wrong, and 20% of negative tests are wrong.) Suppose you are selected for a random screening, and you test positive. What's the chance that you have the disease?



We want $\mathbb{P}[\text{have disease} \mid \text{test positive}]$. For brevity, let's let

$$D = [\text{have disease}] \quad \text{and} \quad T^+ = [\text{test positive}].$$

In other words, we want

$$\mathbb{P}[D \mid T^+]$$

By the definition of conditional probability,

$$\mathbb{P}[D \mid T^+] = \frac{\mathbb{P}[T^+ \cap D]}{\mathbb{P}[T^+]} \quad (4)$$

We will compute the numerator and denominator separately:

- (Numerator) By the multiplication rule for conditional probability ($\mathbb{P}[A \cap B] = \mathbb{P}[A|B] \mathbb{P}[B]$), we have

$$\mathbb{P}[T^+ \cap D] = \mathbb{P}[T^+ \mid D] \mathbb{P}[D] = (.8)(.01) = 0.008.$$

We've now computed the numerator of Eq. (4).

- (Denominator) By the law of total probability,

$$\begin{aligned} \mathbb{P}[T^+] &= \underbrace{\mathbb{P}[T^+ \mid D] \mathbb{P}[D]}_{=0.008 \text{ by ??}} + \underbrace{\mathbb{P}[T^+ \mid D^c] \mathbb{P}[D^c]}_{=(.1)(.99)=0.099} \\ &= (.8)(.01) + (.1)(.99) \\ &= 0.008 + 0.099 \end{aligned}$$

Finally, having done the necessary work, we can now plug our results into Eq. (4):

$$\mathbb{P}[D \mid T^+] = \frac{0.008}{0.008 + 0.099} \approx .075$$

We conclude that if test positive in a random screening, your probability of actually having the disease is about 7.5%. Is this result surprising?

End of Example 40. $\square$